List of organic compounds for analysis:

1. Benzaldehyde
2. Cinnamaldehyde
3. Acetophenone
4. Benzoic acid
5. Cinnamic acid
6. Urea
7. Glucose
8. Aniline
9. Salicylic acid

REASONING

3. Action with sodium bicarbonate:

Carboxylic acids react with Sodium bicarbonate and liberate CO_2. Evolution of carbon dioxide gives brisk effervescence.

$$R\text{-}COOH + NaHCO_3 \longrightarrow R\text{-}COONa + CO_2 \uparrow + H_2O$$

4. Action with Borsches reagent:

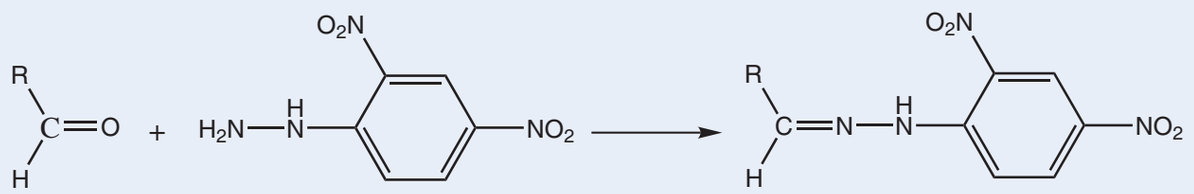
Borsches reagent is prepared by dissolving 2,4-dinitrophenylhydrazine in a solution containing methanol and little of conc sulphuric acid.

Aldehydes and ketones react with borsches reagent to form yellow, orange or red precipitate (dinitro phenylhydrazone)

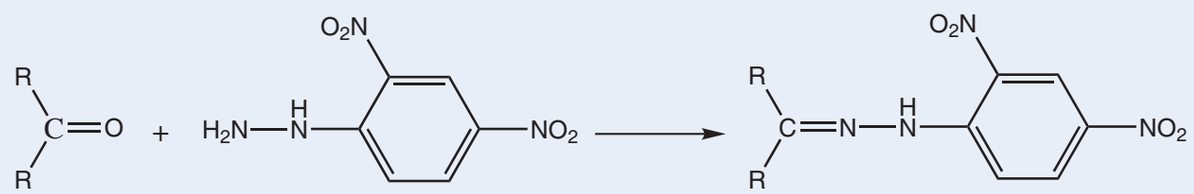
Aliphatic carbonyl compounds give deep yellow precipitate.

Aromatic carbonyl compounds give red precipitate.

2,4-dinitrophenyl hydrazine can be used to qualitatively detect the carbonyl group of an aldehyde or ketone. A positive result is indicated by the formation of an yellow or orange-red precipitate of 2,4-dinitrophenyl hydrazone.



Aldehyde + 2,4 dinitrophenylhydrazine → Aldehyde 2,4 dinitrophenylhydrazone (Yellow or orange)



Ketone + 2,4 dinitrophenylhydrazine → Ketone 2,4 dinitrophenylhydrazone (Yellow or orange or red)

organic analysis.indd 260 22-12-2022 19:48:33